Max Ernst

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PROFESSIONAL SUMMARY

Robotics engineer with dual degrees in Computer Science and Robotics from the University of Michigan, specializing in computer vision, autonomous systems, and human-robot interaction. Experienced in full-stack robotics development from low-level hardware integration to AI-powered perception pipelines, with applied work spanning remote robotic control, DARPA-funded research, and autonomous navigation. Passionate about building intelligent systems that operate reliably in real-world environments.

EDUCATION

University of Michigan — Ann Arbor, MI
Bachelor of Science in Engineering (B.S.E.) – Computer Science — May 2024
Bachelor of Science in Engineering (B.S.E.) – Robotics — May 2024

Relevant Coursework: 3D Robot Perception, Computer Vision, SLAM and Navigation, Autonomous Vehicles, Robot Operating Systems, Machine Learning, Linear Algebra

EXPERIENCE

Robotics Program Director — Bellevue, WA
StemSci — June 2026 - Ongoing

- Heading multiple robotics programs across StemSci and Steamoji, spanning StemSci's nonprofit high school research initiative and Steamoji's VEX V5/IQ highly competitive programs
- Leading student research projects at StemSci, guiding teams from research design through conference presentation and publication
- Recruit, train, and delegate mentors across teams, managing curriculum and coaching quality while running the fabrication lab and facility management
- Building parent/student app and a data-driven system to track team performance and surface trends for coaching strategy

Robotics Engineer (Contract) — Bothell, WA
Dopl Technologies — January 2026 - May 2026

- Hired on fixed-term contract to bring medical grade Kuka robotic arm to Alpha completion phase
- Designed and deployed 1-1 telerobotic control pipeline of a Kuka LBR Med 800 incorporating intelligent nullspace navigation, IEC 80601 and ISO 14971 compliant safety controls, haptic feedback of patient body using Haply Inverse3, and 3D patient surface capture with RealSense RGBD cameras
- Integrated 3-DOF force sensing to enable patient-compliant contact control and safe body interaction during ultrasound exams
- Delivered customer friendly interface using real-time streaming of robot data, including joints, tool pose, force/torque sensor data, and controller state for interpretation and visualization from downstream components, and for later cloud-based data recording and analysis

Corso Group Research Associate — Ann Arbor, MI
University of Michigan, College of Engineering — June 2024 - August 2025

- Conducted research in computer vision and artificial intelligence, specializing in dataset curation, measurement, and evaluation for object detection and segmentation systems
- Developed methodology to analyze completeness and quality of dataset by evaluating CLIP-encoding patterns and cross referencing batch-trained detectors to detect unlabeled items

- Collaborated on DARPA-funded project deploying a mobile robot that fused LiDAR-based SLAM for localization and mapping with a camera-driven vision-language model and tree graph architecture to semantically evaluate and navigate real-world environments
- Partnered with defense contractor Third Insight to develop an RGB/IR YOLO-based military object detection and rangefinding system for multi-modal sensing environments.
- Designed and executed data curation pipelines to assess dataset completeness, balance class distributions, and surface unlabeled or underrepresented samples, directly improving downstream model quality

Vulcan Autonomous Wheelchair Ann Arbor, MI
University of Michigan, College of Engineering June 2023 - September 2023

- Rebuilt and restored core hardware systems by repairing and soldering damaged electronics and reconstructing critical structural components of the autonomous wheelchair platform
- Revived the wheelchairs software by improving YOLO based topological and LiDAR based metrical mapping systems to to increase reliability and enable better socially-aware navigation in hospital environments

Software Engineering Intern Plymouth, MI
Robotire May 2022 - August 2022

- Refactored and expanded upon RGBD wheel well depth analysis using Zivid 3D cameras to accurately identify lugnut locations for automated tire-changing robots
- Developed VIN-based vehicle identification database and lookup system to retrieve vehicle specifications and support automated service workflows, with vehicle data stored and accessed via AWS S3
- Designed and implemented comprehensive utilities testing framework achieving 98% code coverage, identifying performance inefficiencies and improving system reliability

PROJECTS

Autonomous SLAM Navigation Robot

- Built a fully autonomous ground robot capable of simultaneous localization and mapping (SLAM) using fused LiDAR and camera data for environment perception and self-localization
- Integrated camera-based color detection with the SLAM-built map to identify and navigate toward color-coded targets within a maze environment
- Implemented autonomous path planning through the maze using the SLAM-generated occupancy map, enabling the robot to navigate from start to goal without human intervention

Kasparobot AI-enabled Chess Robot

- Designed and built an autonomous chess-playing robotic system integrating computer vision, chess AI, and robotic manipulation
- Implemented a YOLO-based piece detection pipeline with Hough line homography calibration to convert camera observations of the board into machine-readable FEN game states
- Integrated the Stockfish chess engine to generate optimal moves and translate AI outputs into robotic action commands
- Developed a ROS-based control interface to convert chess engine decisions into precise robotic arm movements for piece manipulation

3D Encoding Explorer

- GUI to encode datasets using CLIP and squash encodings to 3D explorable space to highlight emergent physical manifestations of image differences
- Offers tools to visualize images by class, centroids of classes, and difference of specific image to expected centroid based on class labelings

- Allows a user to explore “eigen-images”, or images that represent the direction of maximum variance for its class

Pokemon Design Generator Using GAN

- Developed a generative adversarial network (GAN) for image synthesis trained on a curated dataset of Pokémon designs
- Built and trained the model using PyTorch on the University of Michigan HPC cluster, managing training jobs with SLURM
- Implemented extensive image augmentation pipelines to expand dataset diversity and improve generator performance and training stability

Stable Diffusion Image Generator

- Implemented a diffusion-based training and inference pipeline for text-conditioned image generation consisting of a U-Net architecture, text encoder, and latent image decoder
- Trained the model on the Flickr8k dataset using PyTorch with CUDA acceleration on a local NVIDIA GPU

SKILLS

Programming Languages: Python, C++, C, Javascript, Java, Typescript, HTML, CSS, R, Matlab, SQL
Hardware: Kuka Robots, LiDAR Sensors, Computer construction, Camera integration, Pressure Sensors, IMUs, Soldering
Tools: OpenCV, FiftyOne, YOLO, CLIP, GroundingDINO, SegmentAnything, Pytorch, Tensorflow, CUDA, Llama, Scikit-Learn, Matlab, ROS, Git, Jira
Personal & Soft Skills: Agile workflow, SCRUM management, Investor outreach, Public speaking, Presentations, Technical Communication, Resourceful, Fast Learner